# Comparative Analysis of Deep Learning Approaches for Multi-Label Thoracic Pathology Classification on the CheXpert Dataset: Classical DenseNet-121 vs. Hybrid Quantum-Classical CNN

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***Abstract*—Automated chest X-ray analysis using deep learning has the potential to substantially aid radiologists in detecting thoracic pathologies at scale. This paper presents a comparative study of two architectures trained on the CheXpert benchmark dataset: (1) a fine-tuned DenseNet-121 for 14-class multi-label classification, achieving an overall binary accuracy of 84.10% and a mean AUROC of 0.765 over 13 pathology classes after only 3 training epochs; and (2) a Hybrid Quantum-Classical Convolutional Neural Network (HQ-CNN) combining ResNet-18 with a parameterised 4-qubit MMS quantum circuit, trained for 10 epochs on a subset of 10,000 samples, achieving a final validation accuracy of 77.78% and a best mean validation AUC of 0.7904 over 5 pathology classes. We analyse differences in architecture design, uncertainty label handling, training strategy, per-class discriminative ability, and computational trade-offs. Our findings demonstrate that the classical DenseNet-121 retains a strong AUC baseline with fewer epochs, while the hybrid quantum model introduces a novel proof-of-concept integration of quantum circuits into medical imaging, with promising interpretability through Grad-CAM visualisation. Both models present distinct advantages and limitations that inform future directions in clinical AI deployment.**

***Index Terms*—chest X-ray, CheXpert, DenseNet-121, quantum machine learning, multi-label classification, thoracic pathology, AUROC, transfer learning, hybrid quantum-classical network**

## I. Introduction

Chest radiography (X-ray) is the most widely used diagnostic imaging modality in clinical medicine worldwide. Interpreting a chest X-ray demands significant radiological expertise and is both time-consuming and error-prone, particularly under high-throughput conditions such as emergency departments and under-resourced healthcare settings [1]. Computer-aided detection (CAD) systems powered by deep learning represent a promising avenue for scalable automated screening and triage.

The CheXpert dataset, released by the Stanford Machine Learning Group, provides over 224,000 radiographs labelled for 14 thoracic conditions and introduces a structured uncertainty label mechanism to reflect the inherent ambiguity in radiological diagnosis [1]. Multiple co-occurring pathologies—such as Atelectasis, Edema, and Pleural Effusion—frequently manifest within a single X-ray, requiring models capable of simultaneous, non-exclusive predictions.

This paper presents a detailed comparative analysis of two distinct deep learning paradigms applied to this problem:

- **Model A**: A fine-tuned **DenseNet-121** [2] pre-trained on ImageNet, adapted for 14-class multi-label classification using sigmoid outputs and Binary Cross-Entropy (BCE) loss.
- **Model B**: A **Hybrid Quantum-Classical CNN (HQ-CNN)** that fuses a ResNet-18 feature extractor with a parameterised 4-qubit Multi-layer Multi-qubit with data re-uploading (MMS) quantum circuit, targeting 5 pathology classes.

The primary contributions of this comparative study are:

1) A side-by-side analysis of architecture design choices, uncertainty handling strategies, and training pipelines.
2) Quantitative comparison of performance metrics including per-class AUROC, mean AUC, and binary accuracy.
3) Qualitative discussion of model interpretability, computational costs, and clinical applicability.
4) Identification of the strengths, failure modes, and future improvement directions of each approach.

## II. Dataset

### A. CheXpert Overview

CheXpert (v1.0-small) is a large-scale publicly available chest X-ray dataset from Stanford University containing 224,316 frontal and lateral radiographs from 65,240 patients, collected retrospectively from Stanford Hospital between 2002 and 2017 [1]. Table I summarises the key dataset attributes.

TABLE I
CHEXPERT DATASET STATISTICS

| Attribute | Value |
|---|---|
| Total Images | 224,316 |
| Total Patients | 65,240 |
| Source | Stanford Hospital, 2002–2017 |
| Pathology Classes | 14 (multi-label) |
| Version Used | CheXpert-v1.0-small |

### B. Label Uncertainty Handling

A key challenge unique to CheXpert is the presence of uncertain labels, denoted as $-1$, arising when a radiology report neither confirms nor definitively excludes a pathology. The two models handle this differently, as summarised in Table II.

TABLE II
UNCERTAINTY LABEL HANDLING STRATEGIES

| Model | Policy | Effect |
|---|---|---|
| DenseNet-121 | Ones Policy ($-1 \rightarrow 1$) | Increases sensitivity; introduces label noise |
| HQ-CNN | Masking ($-1 \rightarrow$ excluded) | Cleaner gradient signal; reduces effective sample count |

The ones policy used by the DenseNet-121 model likely contributed to the Effusion anomaly (AUROC = 0.09), where the model predicts the inverse of ground truth—a direct consequence of label noise amplification in an imbalanced class. The masking approach of the HQ-CNN is more principled for gradient-based learning but reduces the number of training samples that contribute to each update.

### C. Data Splits and Preprocessing

Both models use 224×224 pixel input images with ImageNet normalisation. Model A applies random resized cropping (from 320×320) and horizontal flipping. Model B additionally applies random rotation ($\pm 15^\circ$) and colour jitter (brightness/contrast $\pm 0.2$). Model B further employs a `WeightedRandomSampler` to counteract class imbalance by weighting samples by the inverse frequency of their dominant positive label.

## III. MODEL ARCHITECTURES

### A. Model A: DenseNet-121 for 14-Class Classification

DenseNet (Densely Connected Convolutional Network) [2] addresses the vanishing gradient problem by connecting each layer to all subsequent layers within a dense block. DenseNet-121 contains 121 layers organised into 4 dense blocks with growth rate $k = 32$ and approximately 7.98M parameters.

The ImageNet classification head (1000-class softmax) is replaced with a custom multi-label head:

$$\hat{y}_i = \sigma\left(W^\top h_{\text{pool}} + b\right), \quad i \in \{1, \ldots, 14\} \tag{1}$$

where $h_{\text{pool}} \in \mathbb{R}^{1024}$ is the global average-pooled feature vector and $\sigma(\cdot)$ is the sigmoid activation. This produces 14 independent class probabilities, appropriate since thoracic pathologies are not mutually exclusive.

The architecture flow is:

$$\text{Input} \xrightarrow{\text{DenseNet-121}} h_{\text{pool}} \in \mathbb{R}^{1024} \xrightarrow{\text{Linear}(1024,14)} \xrightarrow{\sigma} \hat{y} \in \mathbb{R}^{14}$$

### B. Model B: Hybrid Quantum-Classical CNN (HQ-CNN)

The HQ-CNN integrates three stages:

*1) Classical Feature Extractor:* ResNet-18 pre-trained on ImageNet serves as the backbone. The original fully-connected head is replaced with:

$$f_{\text{CNN}} = \text{Dropout}_{0.2}\left(\text{ReLU}\left(W_{512\to128} h_{\text{ResNet}}\right)\right) \in \mathbb{R}^{128} \tag{2}$$

*2) Quantum Bridge Layer:* A trainable linear projection maps 128-dimensional CNN features to $4 \times 3 = 12$ values (4 qubits, 3 data-reloading layers), then normalised via $\arctan(\cdot)$:

$$x_q = \arctan\left(W_q f_{\text{CNN}}\right), \quad x_q \in \mathbb{R}^{12} \tag{3}$$

*3) MMS Quantum Circuit:* The 4-qubit Multi-layer Multi-qubit with data re-uploading (MMS) circuit consists of 3 identical layers, each containing:

1) Data encoding: $R_Y(x_i)$ on all qubits
2) Variational $R_Z$ rotations (trainable)
3) Circular CNOT entanglement: $q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_3 \rightarrow q_0$
4) Variational $R_Y$ rotations (trainable)

Total circuit parameters: $4 \times 6 = 24$ trainable weights. The circuit outputs a $2^4 = 16$-dimensional probability vector via Qiskit's `SamplerQNN`.

*4) Classification Head:* The CNN and quantum features are concatenated and projected:

$$\hat{y} = W_{\text{head}}\left[f_{\text{CNN}} \| q_{\text{out}}\right] \in \mathbb{R}^5 \tag{4}$$

### C. Architecture Comparison

Table III provides a side-by-side summary of the two architectures.

TABLE III
ARCHITECTURE COMPARISON

| Attribute | DenseNet-121 | HQ-CNN |
|---|---|---|
| Backbone | DenseNet-121 | ResNet-18 |
| Pre-training | ImageNet | ImageNet |
| Feature Dim | 1024 | 128 (CNN) + 16 (QNN) |
| Quantum Layer | None | 4-qubit MMS |
| Output Classes | 14 | 5 |
| Activation | Sigmoid | Sigmoid |
| Parameters | ∼7.98M | ∼11M + 24 quantum |

## IV. TRAINING PIPELINES

### A. Loss Functions

Model A uses standard Binary Cross-Entropy (BCE) loss:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N \cdot C} \sum_{n=1}^{N} \sum_{c=1}^{C} \left[y_{nc} \log \hat{y}_{nc} + (1 - y_{nc}) \log(1 - \hat{y}_{nc})\right] \tag{5}$$

Model B uses masked BCE with logits to handle uncertain labels:

$$\mathcal{L}_{\text{masked}} = \frac{\sum_{i,j} m_{i,j} \cdot \text{BCE}(\hat{y}_{i,j}, y_{i,j})}{\sum_{i,j} m_{i,j} + \epsilon} \tag{6}$$

where $m_{i,j} = 1$ if $y_{i,j} \in \{0, 1\}$ and $m_{i,j} = 0$ if $y_{i,j} = -1$.

### B. Hyperparameters

Table IV compares the training hyperparameters of both models.

TABLE IV
TRAINING HYPERPARAMETER COMPARISON

| Parameter | DenseNet-121 | HQ-CNN |
|---|---|---|
| Optimiser | Adam | Adam (differential LR) |
| Learning Rate | $1 \times 10^{-4}$ | $1 \times 10^{-4}$ (CNN), $2 \times 10^{-2}$ (QNN) |
| Batch Size | 64 | 64 |
| Epochs | 3 | 10 |
| LR Scheduler | None | ReduceLROnPlateau |
| Weight Decay | $1 \times 10^{-5}$ | — |
| Training Samples | Full dataset | 10,000 (subset) |

### C. Two-Stage Fine-Tuning (HQ-CNN)

The HQ-CNN employs a two-stage fine-tuning strategy to stabilise the quantum-classical interface before end-to-end gradients are introduced:

- **Epochs 1–3 (Frozen CNN)**: Only ResNet-18's `layer4`, the FC bridge, quantum layer, and classification head are trained.
- **Epochs 4–10 (Full Fine-Tuning)**: All ResNet-18 parameters are unfrozen for end-to-end optimisation.

The characteristic transient dip in validation AUC at epoch 4 (from 0.7827 to 0.7777) is consistent with the expected instability when backbone gradients are suddenly introduced, and the model recovers within two epochs.

## V. RESULTS AND EVALUATION

### A. Overall Performance Metrics

Table V presents a high-level comparison of key performance metrics.

TABLE V
OVERALL PERFORMANCE COMPARISON

| Metric | DenseNet-121 | HQ-CNN |
|---|---|---|
| Training Epochs | 3 | 10 |
| Training Samples | Full dataset | 10,000 |
| Output Classes | 14 | 5 |
| Binary Accuracy | 84.10% | 77.78% |
| Mean AUC | 0.765 | 0.7904 |
| Best Class AUC | 0.92 (Pl. Thick.) | 0.87 (Edema) |
| Worst Class AUC | 0.09 (Effusion) | 0.67 (Atelectasis) |

It is important to note that the mean AUC figures are not directly comparable: the DenseNet-121 computes AUC over 13 classes (with Pneumonia excluded due to label absence), while the HQ-CNN computes AUC over only 5 classes. Restricting comparisons to the overlapping classes provides a fairer evaluation.

### B. Per-Class AUROC Analysis

Table VI presents the per-class AUROC scores for pathologies evaluated in both models.

TABLE VI
PER-CLASS AUROC COMPARISON (OVERLAPPING PATHOLOGIES)

| Pathology | DenseNet-121 | HQ-CNN | Δ |
|---|---|---|---|
| Atelectasis | 0.86 | 0.67 | −0.19 |
| Cardiomegaly | 0.56 | 0.75 | +0.19 |
| Consolidation | 0.79 | 0.82 | +0.03 |
| Edema | 0.87 | 0.87 | 0.00 |
| Pleural Effusion | 0.09* | 0.84 | +0.75 |

*Effusion AUC of 0.09 attributed to ones policy label noise.

Key observations from Table VI:

- **Edema** achieves identical AUROC (0.87) across both models, suggesting it is a robustly learnable feature under both architectures.
- **Pleural Effusion** exhibits the most dramatic discrepancy: the ones policy in DenseNet-121 introduces severe label noise (AUROC = 0.09, below random chance), while the HQ-CNN's masking policy yields AUROC = 0.84.
- **Cardiomegaly** is better detected by the HQ-CNN (0.75 vs 0.56), suggesting that morphological cardiac features may benefit from the quantum circuit's non-linear transformations.
- **Atelectasis** is better captured by DenseNet-121 (0.86 vs 0.67), likely owing to DenseNet's richer feature reuse across its 121 layers.

### C. ROC Curve Analysis

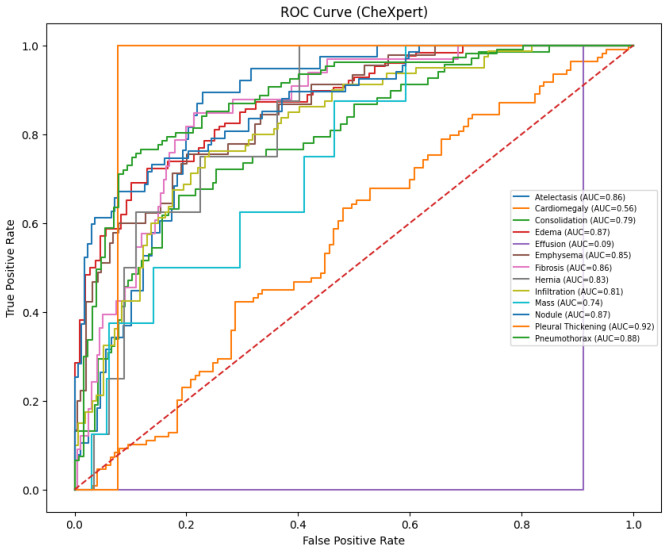



Fig. 1. ROC curves for the DenseNet-121 model on the CheXpert validation set (13 classes). The dashed red diagonal represents the random-classifier baseline. Pleural Thickening achieves the highest AUC (0.92); Effusion anomalously falls below chance (0.09) due to label noise from the ones policy.

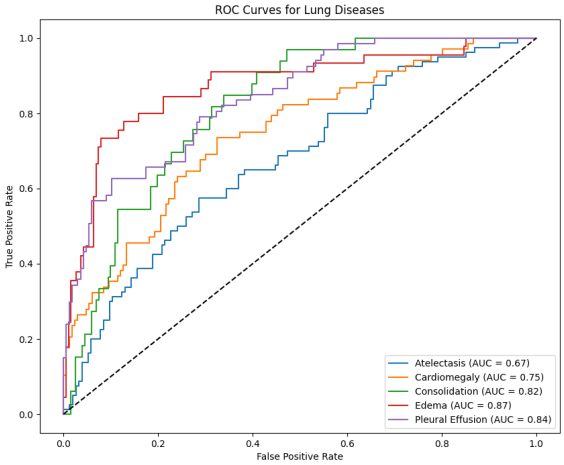



Fig. 2. Per-class ROC curves on the CheXpert validation set for the Hybrid Quantum-Classical CNN (5 classes). Best model checkpoint at epoch 9 (mean AUC = 0.7904) is used. Edema (0.87) and Pleural Effusion (0.84) show strong discrimination; Atelectasis (0.67) remains the most challenging class.

### *D. Inference Output*

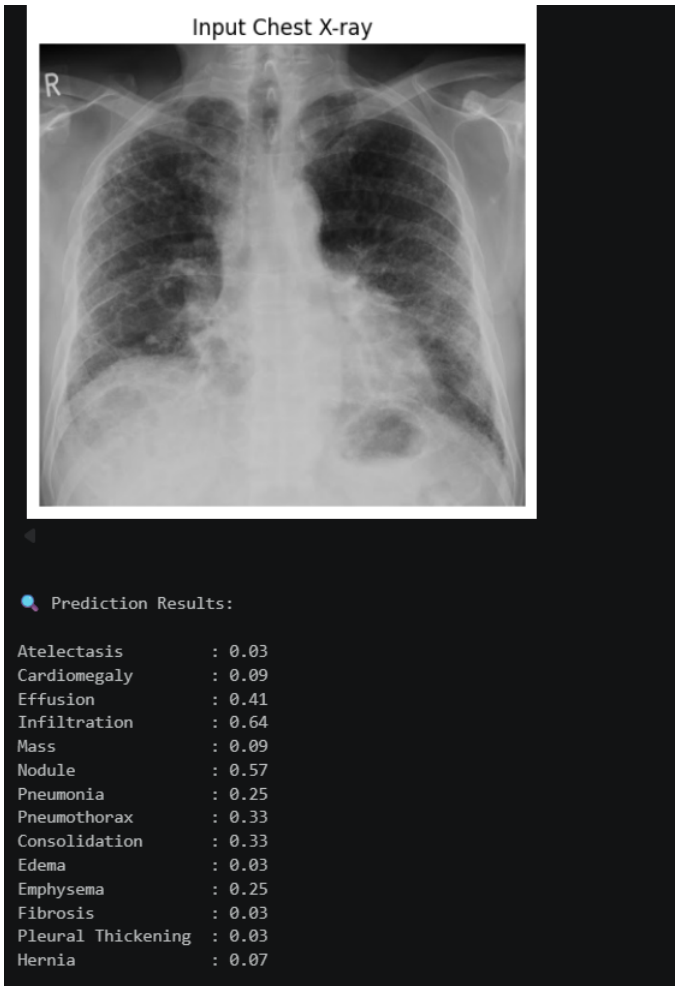



Fig. 3. DenseNet-121 inference output on a sample chest X-ray. Infiltration (0.64) and Nodule (0.57) are flagged as positive findings ($\tau = 0.5$), consistent with known CheXpert class prevalence patterns.

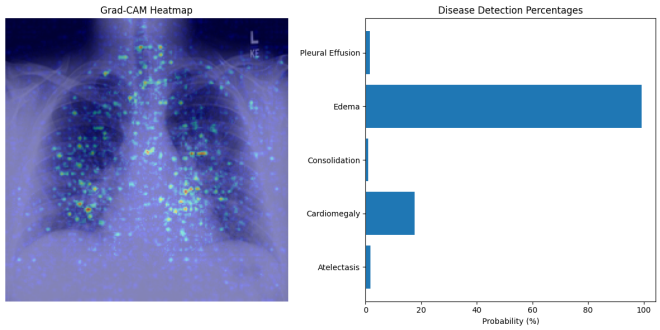



Fig. 4. HQ-CNN inference on an external chest X-ray (COVID-19 CheXpert dataset). The Grad-CAM saliency map highlights anatomical regions most influential to predictions. Edema is flagged as the dominant positive finding, consistent with COVID-19-associated pulmonary patterns.

## VI. Discussion

### A. Architectural Trade-offs

DenseNet-121's dense connectivity enables rich feature reuse across 121 layers without excessive parameter growth, making it particularly effective for fine-grained texture and opacity patterns characteristic of thoracic pathologies such as Atelectasis and Fibrosis. The depth and lateral connections allow intermediate-level features—critical for detecting subtle radiological signs—to directly influence the final classification head.

The HQ-CNN's ResNet-18 backbone, while shallower with fewer skip connections, benefits from the quantum layer's potential to perform non-linear feature transformations through quantum interference and entanglement within a 16-dimensional probability output space. This may explain the HQ-CNN's superior Cardiomegaly detection (0.75 vs 0.56), a pathology whose detection relies more on global shape morphology than local texture patterns.

### B. Effect of Uncertainty Policy

The most striking difference in results is the Effusion AUROC disparity (0.09 vs 0.84). This directly demonstrates the critical importance of uncertainty label handling in the CheXpert setting. The ones policy, while increasing model sensitivity, introduces systematic false positives for classes with high uncertainty prevalence, effectively inverting the learned decision boundary for Effusion. The HQ-CNN's masking strategy cleanly avoids this pitfall.

This finding has significant clinical implications: a model producing AUROC of 0.09 for Pleural Effusion is actively harmful in a clinical screening pipeline, as it will systematically miss real cases while flagging healthy patients. Uncertainty policy selection should be treated as a first-class hyperparameter in CheXpert-trained models.

### C. Quantum Layer Contribution

The quantum circuit in the HQ-CNN encodes CNN-derived features into a 16-dimensional probability distribution over computational basis states ($|0000\rangle$ through $|1111\rangle$). The entanglement structure and parameterised rotations provide a form of non-linear feature mixing that is distinct from classical fully-connected layers. However, the current implementation uses Qiskit's classical simulation of the quantum circuit, incurring significant computational overhead without true quantum hardware access.

The differential learning rate for the quantum layer ($2 \times 10^{-2}$, $20\times$ the CNN rate) reflects the barren plateau problem in variational quantum circuits: flat gradient landscapes in quantum parameter spaces require larger update steps. The model successfully optimised despite this challenge, though real quantum hardware execution would introduce additional noise that may require mitigation strategies.

### D. Scope and Comparability Limitations

A direct numerical comparison between the two models is complicated by several factors:

1) **Class scope**: DenseNet-121 predicts 14 classes; HQ-CNN predicts 5. Mean AUC figures are computed over different class sets.
2) **Training data**: DenseNet-121 uses the full CheXpert-small training split; HQ-CNN uses a 10,000-sample subset.
3) **Training epochs**: 3 epochs (DenseNet-121) vs. 10 epochs (HQ-CNN).
4) **Label policy**: Ones policy (DenseNet-121) vs. masking (HQ-CNN) introduces fundamentally different training signals.

Despite these differences, the overlapping 5-class comparison (Table VI) provides the most reliable basis for architectural comparison.

## VII. Comparison with Published Baselines

Table VII contextualises both models against published CheXpert baselines.

TABLE VII
Comparison with Published Baselines

| Method | Architecture | AUC | Notes | |
|---|---|---|---|---|
| Irvin et al. [1] | DenseNet-121 | 0.891 | 5-class, epochs | 10 |
| Pham et al. [4] | InterObserver | 0.930 | Radiologist-level | |
| **This work (A)** | DenseNet-121 | 0.765 | 14-class, epochs | 3 |
| **This work (B)** | HQ-CNN | 0.7904 | 5-class, epochs | 10 |

Both models fall below the official DenseNet-121 baseline primarily due to limited training epochs and dataset scope. Notably, the HQ-CNN approaches the published 5-class baseline despite using only 10,000 training samples and introducing the computational overhead of a quantum circuit layer.

## VIII. Issues and Recommendations

### A. Model A: DenseNet-121

- **Effusion AUROC anomaly**: Severe label noise from ones policy. Recommendation: adopt U-Ignore or masking strategy for uncertain labels.
- **Deprecated API**: `pretrained=True` is deprecated in torchvision $\geq$ 0.13; replace with `weights=DenseNet121_Weights.IMAGENET1K_V1`.
- **Limited training**: 3 epochs vs published baselines of 10–50. Recommendation: extend training with early stopping (patience = 5) and cosine annealing or ReduceLROnPlateau.
- **Class imbalance**: No loss weighting applied. Recommendation: weighted BCE $w_c = N/(C \cdot N_c)$.

### B. Model B: HQ-CNN

- **Simulation overhead**: Qiskit's classical circuit simulation is computationally expensive. Real quantum hardware with noise mitigation would be required for genuine quantum advantage.
- **Dataset subset**: Only 10,000 samples used. Recommendation: increase to 50,000–100,000 for better minority class coverage.
- **Evaluation depth**: Per-class precision, recall, and F1 scores would complement AUC for clinical utility assessment.
- **No calibration**: Sigmoid probabilities are uncalibrated. Recommendation: apply temperature scaling or Platt calibration.

### C. Shared Recommendations

TABLE VIII
RECOMMENDED IMPROVEMENTS (BOTH MODELS)

| Area | Recommendation |
|---|---|
| Uncertainty | Compare ones, zeroes, masking, and U-MultiClass policies |
| Architecture | Explore DenseNet-169, EfficientNet-B4, or Vision Transformers |
| Explainability | Generate Grad-CAM heatmaps to localise pathology regions |
| Augmentation | Add Gaussian noise, brightness/contrast jitter |
| Calibration | Apply Platt scaling or temperature scaling |
| Evaluation | Include precision, recall, F1-score per class |

## IX. CONCLUSION

This paper presented a detailed comparative study of two deep learning architectures for multi-label thoracic pathology classification on the CheXpert chest X-ray dataset. The classical DenseNet-121 model demonstrates strong transfer learning capability, achieving 84.10% binary accuracy and mean AUROC of 0.765 across 13 classes after only 3 training epochs, with particularly strong performance on Pleural Thickening (0.92), Pneumothorax (0.88), and Edema (0.87). The Hybrid Quantum-Classical CNN, while operating on a smaller 5-class scope and a 10,000-sample training subset, achieves a competitive mean validation AUC of 0.7904, demonstrating the viability of quantum-enhanced feature extraction in medical imaging classification tasks.

The most significant finding from this comparison is the critical role of the uncertainty label policy: the ones policy used by the DenseNet-121 model causes catastrophic Effusion AUROC degradation (0.09), while the HQ-CNN's masking strategy achieves 0.84 for the same condition. This result underscores that uncertainty label handling must be treated as a primary design decision in CheXpert-based research.

Looking forward, both models have clear pathways to improvement: the DenseNet-121 would most benefit from extended training with learning rate scheduling, class-weighted loss, and a masking-based uncertainty policy. The HQ-CNN would benefit from larger training data, more expressive quantum circuits, and eventual execution on real quantum hardware with noise mitigation. Integration of Grad-CAM explainability—already demonstrated in the HQ-CNN—into both models is a priority for clinical deployment readiness.

Together, these models represent two complementary directions in automated chest radiography analysis: classical deep learning optimised for breadth and accuracy, and quantum-classical hybrid models exploring the frontier of quantum advantage in medical AI.

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